

Generalizable Lithographic Hotspot Detection Using Asynchronous Meta-Learning with Only One Shot

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Abstract—With integrated circuits shrinking in feature size, layout printability has become increasingly challenging, making lithographic hotspot detection ever-crucial in computer-aided design (CAD) flows. In recent years, numerous studies have explored deep learning to detect lithographic hotspots, offering promising results. However, neural networks can easily be biased and overfit when lacking sufficient training data, especially in the CAD domain. A generalizable DL-based hotspot detector should learn the genuine lithography principle and ensure consistent accuracy across layouts from various designs at the same technology node, regardless of their varying design styles. However, we find that existing convolutional neural network (CNN)-based hotspot detectors fail to generalize to different circuit layouts other than the design it has been trained for. To this end, we propose a few-shot learning-based framework for generalizable CNN-based hotspot detection. We develop a meta-learning scheme that asynchronously updates the CNN feature extraction and classification component to obtain a meta-initialized model that can quickly adapt to new designs using as few as one training layout clip. We propose a layout topology-based sampling strategy for few-shot adaptation to enhance generalization stability. Experimental results on ICCAD 2012 and 2019 datasets show that our framework enables superior generalization capabilities than prior arts on unseen new designs.

Index Terms—Lithographic hotspot detection, generalizability, few-shot learning

I. INTRODUCTION

With the shrinkage of feature sizes in advanced technology nodes, integrated circuit layouts are susceptible to printing defects, making lithographic hotspot detection a crucial part of the design-for-manufacturability (DFM) flow. Conventional lithography simulation achieves accurate hotspot detection at the expense of high computation costs. Pattern matching-based methods expedite the detection process by comparing layout patterns with existing hotspots but fail to identify hotspots unseen before. Recently, with advancements in deep learning (DL), researchers have proposed DL-based solutions that demonstrate promising results in hotspot detection.

Most existing DL-based works focus on improving hotspot detection accuracy. For example, Yang et al. [1] proposed a DCT-based layout feature representation for hotspot detection. Geng et al. [2] introduced attention modules in the DL model, achieving an end-to-end hotspot detection flow.

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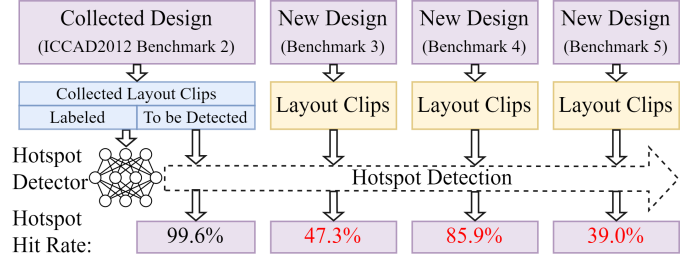


Fig. 1 Detection accuracy of SoTA attention-based CNN hotspot detector [2] when validated on various layout designs using the same 28 nm technology node. *Note that ICCAD 2012 Benchmark 1 is developed using a different 32 nm technology node and thus is not tested here.*

Jiang et al. [3] proposed a Bayesian optimization-based neural architecture search scheme to automatically design high-performance convolutional neural networks (CNNs). Shao et al. [4] integrated a learned lithography simulator with an object detection backbone to extract layout features, thereby improving detection performance. In addition to accuracy enhancements, other studies explore orthogonal directions in DL-based hotspot detection, including inference speed-up [5], [6], data scarcity [7], [8] and imbalance [9]–[11], robustness and security [12]–[14], and interpretability [15], [16]. Other works detect multiple hotspots on full layouts instead of partitioned layout clips [17], [18].

However, despite the verified efficiency of CNNs for hotspot detection as demonstrated by their accuracy when trained and validated using a certain circuit design, it has never received deliberation whether they have learned the genuine lithography principle rather than a biased design style pertaining to a specific design. Given that the CNNs are notorious for having overfitting issues and are easy towards biased learning [19], in this work, we seek to answer the following question:

Can DL-based lithographic hotspot detectors generalize to unseen new designs different from their trained design at the same technology node?

We use a state-of-the-art (SoTA) attention-based CNN hotspot detector from [2] as our motivating case study, and we examine its detection accuracy on various circuit designs from the ICCAD 2012 dataset that uses the same 28 nm technology node. Specifically, we use the training dataset from benchmark 2 for training and the rest of the layout clips from benchmark 2 to 5 for testing, as shown in Fig. 1. We witness a drastic

drop between detection accuracy on benchmark 2 and that on benchmark 3, 4, and 5, despite their shared technology node used in layout design, indicating severe generalization limitations of existing hotspot detection CNNs.

Several studies [20]–[23] have explored transfer learning or active learning with a certain budget for CNN-based hotspot detection when design environments and parameters change. However, they rely heavily on significant amounts of labeled data and are sensitive to the initial model, which can significantly affect the final performance. In the case of CAD, large amounts of quality labeled data are not always accessible.

To this end, we propose a training framework for generalizable CNN-based hotspot detection on unseen new designs using a minimum additional budget. Specifically, we propose an asynchronous meta-learning scheme that obtains a generalizable meta-initial model that can adapt to new designs using few-shot learning. Asynchronous updates are applied to CNN’s feature extraction and classification component, alleviating the overfitting of specific feature representations of individual designs. We further develop a layout topology-based sampling strategy to select the most representative layout patterns in the new design for a few-shot adaptation of the first stage meta-initialized model. Our experimental results demonstrate the efficiency of our framework, showcasing both stable and high-performance hotspot detection across designs. Our main contributions are as follows:

- We investigate for the first time the generalizability of DL-based lithographic hotspot detectors on different circuit designs at the same technology node.
- An asynchronous meta-learning framework that trains a CNN hotspot detector that can generalize to unseen new designs using as few as one-shot adaptation.
- A layout topology-based sampling strategy for one-shot adaptation that enables the most stable generalization.
- Extensive experimental results on the advantages of our proposed training framework for generalizable DL-based hotspot detection compared to all baseline solutions.

II. PRELIMINARIES AND PROBLEM DEFINITION

A. Meta-learning

Meta-learning refers to a variety of techniques that learn knowledge with accumulated data of multiple tasks from prior experiences and then quickly adapt to new tasks with few data points. In [24], Bai et al. categorize existing studies in meta-learning into three types, including metric-based, memory-based, and learning-based methods. Among them, MAML [25] is a representative learning-based algorithm designed to learn a cross-task meta-initial parameter θ that enables rapid adaptation to new tasks.

In MAML, the dataset $\mathcal{D}$ comprising all tasks is first divided into two non-overlapping meta sets: $\mathcal{D}^{train}$ for meta-training, and $\mathcal{D}^{val}$ for validation. $\mathcal{D}^{train}$ is further partitioned into the *Support set* $\mathcal{S}^{train}$ and the *Query set* $\mathcal{Q}^{train}$ to facilitate the *Inner loop* and the *Outer loop* of meta-training. The inner loop refines the initial parameter θ using $\mathcal{S}_i^{train}$ for

each task i , yielding task-specific intermediate parameters θ_i . The outer loop then refines the meta-initial parameter with $\mathcal{Q}^{train}$ to enhance cross-task generalization. After meta-training, adaptation and evaluation are conducted using $\mathcal{S}^{val}$ and $\mathcal{Q}^{val}$, respectively, sampled from $\mathcal{D}^{val}$.

B. Evaluation Metrics and Problem Formulation

We formulate lithographic hotspot detection as a binary classification problem. Given a set of labeled layout clips $\mathcal{D}^{train} = \{\mathcal{D}_1, \mathcal{D}_2 \dots \mathcal{D}_N\} = \{(X_i, Y_i)\}_{i=1}^N$ from N circuit designs, our proposed framework aims to learn a model f with an initial parameter θ such that $Y = f_\theta(X)$ and can quickly adapt to a new design using only K sampled layout clips for training. We adopt the same metrics used in prior lithographic hotspot detection work as follows.

Definition 1 (Hotspot Hit Accuracy, *Acc*). The ratio of correctly classified hotspot clips among all the actual hotspots.

Definition 2 (False Alarm Rate, *FAR*). The ratio of clips misclassified as hotspots among all the actual non-hotspots.

Definition 3 (F1 Score, *F1*). The harmonic mean of *Acc* and *Precision*, where *Precision* denotes the ratio of correctly identified hotspots among the total number of predicted hotspots. *F1* comprehensively reflects *Acc* and *FAR* performance.

$$F1 = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (1)$$

Problem 1 (Generalizable CNN-based Lithographic Hotspot Detection). Given a set of layout clips from collected designs that contain hotspots and non-hotspots, train an initial model that can generalize to both the collected designs and unseen new designs at the same technology node using K -shot (i.e., $K = 1$) adaptation that achieves the maximum *Acc* and *F1* and the minimum *FAR*.

III. PROPOSED METHOD

A. Overview

In this paper, we aim to enhance the generalizability of CNN-based lithographic hotspot detection across various unseen new designs at the same technology node as the trained design, despite their different design styles. Specifically, we propose an asynchronous meta-learning-based framework to train an initial model that can quickly adapt to new designs with a minimum budget, i.e., using one-shot learning with a sampling strategy.

Fig. 2 provides an overview of our proposed framework. Our generalizable hotspot detection process begins with a collection of layout clips and their corresponding labels derived from various designs to construct the meta-training dataset $\mathcal{D}^{train}$. A randomly initialized hotspot detection model is trained on $\mathcal{D}^{train}$ for multiple epochs using the proposed asynchronous meta-learning algorithm to obtain the meta-initial parameter, θ . Following this training phase, the meta-initialized model is adapted to new designs using a sampled set of K layout clips per category from the new design, which is selected by our proposed layout topology-based sampling method.

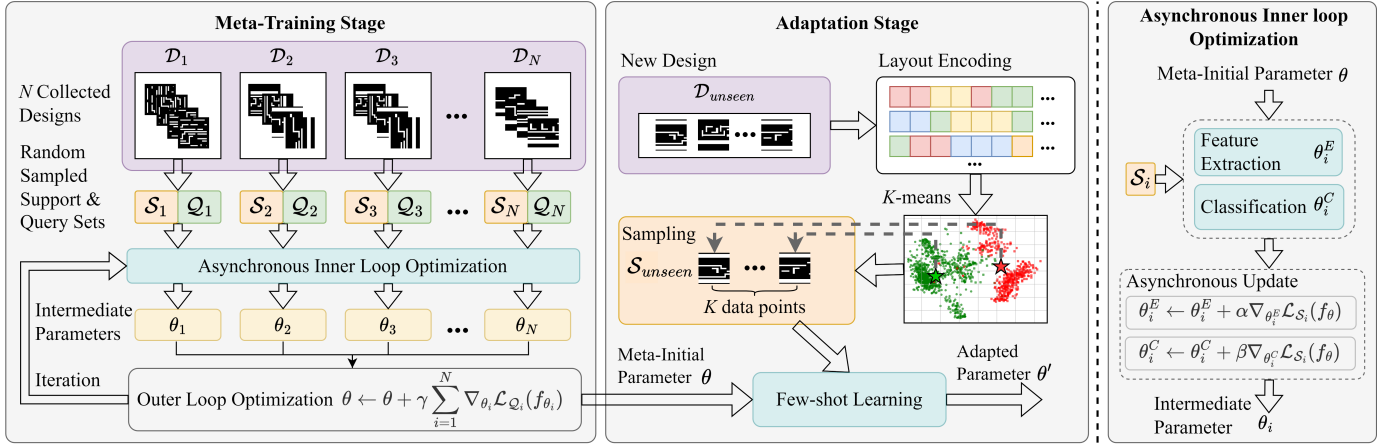


Fig. 2 Illustration of our proposed asynchronous meta-learning framework for generalizable DL-based lithographic hotspot detection using few-shot adaptation.

B. Asynchronous Meta-training

A meta-learning framework incorporates inner loop optimization and outer loop optimization, which update the network parameters based on layout clips from individual designs and all collected designs, respectively. However, the subtle and complex lithography principle is hard to learn, whereas, on the contrary, the extreme bias of design styles of the small support sets (i.e., 2 samples) from a specific design can easily overtake the CNN's learning directions. To suppress overfitting on the design styles of individual layout circuits and gradually converge to the common lithography rules shared among different designs, we constrain the learning capacity of the CNN's feature extraction component when learning each specific design in the inner loop.

We propose an asynchronous meta-training method such that different learning rates are applied to the feature extraction and classification components within each inner loop for each design. Specifically, the feature extraction component is updated with a smaller learning rate, α , to prevent overfitting on individual design features, and the classification component is updated with a relatively larger learning rate, β , to facilitate discrimination capabilities. During the outer loop, when updating with a larger batch size of the query set, both feature extraction and classification components are updated using the same learning rate, γ , to enhance CNN's overall generalization across designs.

1) *Inner loop optimization*: Given a set of collected layout clips $\mathcal{D} = \{\mathcal{D}_1, \mathcal{D}_2 \dots \mathcal{D}_N\}$ from N designs, support sets $\mathcal{S}_i$ and query sets $\mathcal{Q}_i$ are randomly sampled for each design $\mathcal{D}_i$. The intermediate parameters θ_i are obtained through asynchronous inner loop optimization, formulated as follows:

$$\theta_i^E \leftarrow \theta_i^E + \alpha \nabla_{\theta_i^E} \mathcal{L}_{\mathcal{S}_i} \quad (2)$$

$$\theta_i^C \leftarrow \theta_i^C + \beta \nabla_{\theta_i^C} \mathcal{L}_{\mathcal{S}_i} \quad (3)$$

Equation 2 and Equation 3 represent the asynchronous update for CNN's feature extraction component θ_i^E and classification component θ_i^C , using different learning rates α and

β . We adopt the same loss function $\mathcal{L}$ as used in [2] in our training framework, where a prediction and metric loss combine to provide an enhanced hotspot detection accuracy, as expressed in Equation 4, Equation 5, and Equation 6. Here, the metric loss optimizes the distances between the embedding vectors $f_{\theta^E}(x)$ of layout clip x and other layouts from the same (x^+) and different (x^-) categories in the batch. M is a positive constant.

$$\mathcal{L} = \mathcal{L}^{pred} + \mathcal{L}^{metric} \quad (4)$$

$$\mathcal{L}^{pred} = - \sum y \log(f_{\theta}(x)) + (y - 1) \log(1 - f_{\theta}(x)) \quad (5)$$

$$\mathcal{L}^{metric} = \sum \max(0, M + \|f_{\theta^E}(x) - f_{\theta^E}(x^+)\|_2^2 - \|f_{\theta^E}(x) - f_{\theta^E}(x^-)\|_2^2) \quad (6)$$

2) *Outer loop optimization*: After the inner loop optimization, we obtain a set of intermediate parameters, denoted as $\{\theta_1, \theta_2 \dots \theta_N\}$, for each design. Subsequently, gradients of each θ_i are computed based on their corresponding query sets $\mathcal{Q}_i$. These gradients are then aggregated to update the meta-initial parameter θ during the outer loop optimization using learning rate γ , as shown in Equation 7:

$$\theta \leftarrow \theta + \gamma \sum_{i=1}^N \nabla_{\theta_i} \mathcal{L}_{\mathcal{Q}_i}(f_{\theta_i}) \quad (7)$$

Note that in the outer loop optimization, gradients are computed based on a set of design-specific parameters θ_i obtained from the inner loop yet applied to the meta-initial parameter θ . This meta-training procedure ensures the CNN converges toward a common initial parameter, facilitating generalization to new designs that observe a universal lithography principle at the same technology node.

C. Layout Topology-based Few-shot Sampling of New Designs

After meta-training, the CNN takes few-shot adaptive training to generalize to unseen new designs. However, the generalization performance highly depends on the selection of

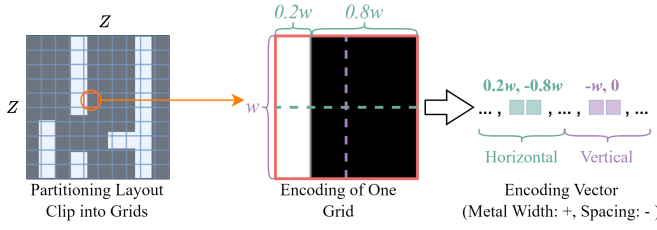


Fig. 3 Illustration of our layout topology-based encoding. A layout clip is partitioned into $Z \times Z$ grids of size $w \times w$. Along horizontal and vertical directions, the lengths of metal widths and spacings are encoded with + and - values, respectively.

layout clips used for adaptation. Training on outlier data can obviously result in inaccurate predictions for the new design. To address this issue, we propose a layout topology-based encoding strategy that assists in sampling the most representative layout clips of the new design. Our layout topology-based encoding represents each layout clip as a fixed-length vector, capturing its structural characteristics, based on which we employ K -means clustering (2-way for hotspots and non-hotspots) that selects the cluster heads and their nearest neighbors as the most representative hotspot and non-hotspot clips in the new design for few-shot adaptation.

1) *Layout topology-based encoding*: Various layout encoding methods have been proposed for hotspot detection, such as concentric circle-based [26], graph-based [6], or density-based [27] representation. However, they either produce variable-length feature vectors or miss out on critical layout information (i.e., using only density information). In the context of new design clips with no accessible ground-truth labels, variable-length or incomplete layout features pose significant challenges for unsupervised layout analysis.

To this end, we propose a layout topology-based encoding method that employs layout topology features for fixed-length feature representation. As illustrated in Fig. 3, for a layout clip of size $R \times R$ with half-pitch p , we partition each clip into the same number of $Z \times Z$ grids, each sizing $w \times w$. Here Z is a user-defined parameter satisfying the constraint $Z > \frac{R}{p}$, such that each grid contains at most one metal wire. We next use horizontal and vertical scanning lines to traverse each grid of the layout clip and capture the geometric information of the metal and spacing within each grid. Specifically, in each grid, a central horizontal line scans from left to right, and a central vertical line scans from top to bottom. Metal and spacing dimensions of the grid are used and encoded as a one-dimensional fixed-length vector, which stacks over all the grids of the layout clip. This scanning approach and encoding ensure full coverage of the layout’s topological features. To differentiate metal and spacing characteristics, we assign metal widths with positive values and spacing widths with negative values. Our fixed-length layout topology-based encoding enables next-step layout comparisons and analyses.

2) *K-means clustering-based few-shot selection*: Since the layout clips of a new design are unlabeled, we employ a 2-way K -means clustering to partition these new clips into two groups in an unsupervised fashion, where the most repre-

sentative clips within each group are selected for a few-shot adaptation. Specifically, our proposed layout topology-based encoding for each new clip is used as input to the K -means clustering algorithm. We select the cluster heads and their nearest $K - 1$ neighbors as the K most representative clips for each layout class. We apply lithography simulation on these sampled layout clips for ground-truth labeling and train them in a one-pass few-shot adaptation for CNN generalization.

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

TABLE 1: Layout clip numbers of each benchmark in ICCAD 2012 & 2019 that use 28 nm technology node.

Dataset		ICCAD 2012				ICCAD 2019			
		B2	B3	B4	B5	B2	B3	B4	B5
Train	HS	174	909	95	26	38	370	46	13
	NHS	5285	4643	4452	2716	5365	5166	4498	2729
Test	HS	498	1808	177	41	278	622	81	20
	NHS	4146	3541	3386	2111	4362	4654	3473	2132

1) *Datasets*: We evaluate the effectiveness of our training framework for generalizable CNN-based hotspot detection on two layout datasets: ICCAD 2012 [28] and ICCAD 2019 [7]. In each dataset, four 28 nm benchmarks (B2 to B5) are employed to train and assess the CNN’s generalizability to new designs developed at the same technology node (Benchmark 1 uses a 32 nm technology node.) Note that layouts in ICCAD 2012 and 2019 are labeled using different lithography systems. Thus, we do not evaluate CNN’s detection generalizability across different datasets. We show in Table 1 the number of hotspot (HS) and non-hotspot (NHS) clips in ICCAD 2012 and 2019 in separate training and testing sets. We use the Test-I dataset in ICCAD 2019 for evaluation and exclude Test-II, which is manually augmented based on Test-I.

In our experiments, we train our CNN hotspot detector on two collected designs and evaluate its generalizability on two new designs. We train on layout collection of the training sets of either (B2 and B4) or (B3 and B5) and evaluate the test set of the rest of the two designs.

2) *Baselines for Comparison*: We compare our proposed framework against four baselines: (1) retraining from scratch using layout clips from new designs, (2) transfer learning the well-trained model on collected designs to new designs, (3) the ADAPT framework proposed by [23] that employs unsupervised Bayesian-based accuracy estimation for initial model selection and active learning with balanced data selection, and (4) meta-initial training as proposed in MAML [25] along with few-shot adaptation. We additionally evaluate our generalization framework that applies synchronous updates for CNN feature extraction and classification components as in standard meta-initial training, denoted as “Ours-” in Table 2.

We randomly sampled K layout clips per category, i.e., hotspot and non-hotspot, for our four baselines in their few-shot adaptation to the base model. Our training framework and “Ours-” use specially designed layout topology-based sampling strategy for few-shot adaptation. In consistency with

TABLE 2: Accuracy comparison across various frameworks incorporating initial training on collected designs, followed by one-pass few-shot adaptation and validation on new designs. In a few-shot adaptation, 1 or 5 layout clips are selected per category from new designs, either randomly (RT, TL, ADP, MAML) or sampled (Ours-, Ours). RT: retraining from scratch, TL: transfer learning, ADP: ADAPT [23]. MAML [25] and ours employ meta-learning for initial training. Ours-: our framework with synchronous meta-learning. Red and blue denote the best and second-best results, respectively.

Old Design	New Design	Method	ICCAD 2012						ICCAD 2019					
			1-shot			5-shot			1-shot			5-shot		
			Acc	FAR	F1	Acc	FAR	F1	Acc	FAR	F1	Acc	FAR	F1
B2 & 4	B3	RT	56.7	52.4	43.7	80.0	79.0	47.8	60.0	59.9	19.7	61.6	62.9	19.5
		TL	58.4	2.3	71.7	66.7	4.0	76.4	82.0	31.3	39.4	83.9	26.9	43.6
		ADP	97.1	27.7	77.2	96.2	24.1	79.1	32.3	9.6	31.7	39.6	11.2	35.5
		MAML	83.9	22.3	73.7	30.1	5.3	42.9	93.1	68.6	26.4	94.5	61.6	28.8
		Ours-	99.7	47.5	65.9	99.8	57.6	63.9	87.3	49.4	31.4	100.0	94.9	22.0
		Ours	95.5	2.1	95.7	96.5	2.8	95.6	99.6	31.2	76.6	99.7	0.3	98.7
	B5	RT	100.0	98.3	3.8	80.0	79.3	3.8	100.0	99.0	1.9	80.0	79.2	1.9
		TL	60.0	0.5	63.9	75.6	3.3	44.1	90.0	13.0	11.4	90.0	12.6	11.8
		ADP	95.1	14.2	20.6	92.2	11.2	24.0	30.0	2.9	13.8	45.0	4.2	15.2
		MAML	75.7	15.0	16.0	73.7	8.0	25.1	100.0	63.5	2.9	89.2	32.9	4.8
		Ours-	100.0	41.4	6.2	100.0	68.4	5.4	100.0	59.2	3.1	100.0	93.1	2.0
		Ours	97.5	1.5	70.9	97.5	1.6	69.6	100.0	7.0	21.3	100.0	3.3	35.7
B3 & 5	B2	RT	80.0	77.4	19.4	20.0	20.0	14.0	80.0	80.0	11.2	60.1	59.9	10.9
		TL	93.7	3.4	84.6	96.2	7.0	75.7	75.8	11.8	42.0	81.3	11.0	46.0
		ADP	94.2	41.1	35.1	90.7	35.7	37.2	70.1	13.1	37.4	65.2	12.6	35.9
		MAML	78.3	29.6	36.9	44.1	8.8	40.6	55.1	16.0	27.1	63.4	29.3	20.3
		Ours-	99.8	58.5	29.1	100.0	88.7	21.3	95.0	31.1	27.8	67.0	28.7	21.7
		Ours	99.4	4.1	75.3	99.4	2.0	91.9	97.5	7.4	62.2	97.4	6.8	64.1
	B4	RT	63.7	64.6	9.1	40.0	40.0	8.8	60.0	60.1	4.4	60.0	54.0	4.8
		TL	75.5	1.1	76.9	84.6	3.8	65.7	64.0	7.2	27.2	79.3	7.0	33.1
		ADP	67.8	23.9	21.7	70.7	26.4	20.9	60.5	7.1	25.9	63.0	7.6	25.8
		MAML	77.7	13.8	35.2	85.1	7.1	53.1	63.4	16.6	14.5	77.0	26.2	11.9
		Ours-	99.4	32.2	24.4	65.3	4.5	52.0	88.9	20.5	16.7	88.9	29.6	13.5
		Ours	96.6	6.6	59.7	95.4	4.7	66.7	95.0	4.0	51.7	97.6	4.9	48.0

common practices in few-shot learning studies, we evaluate $K = \{1, 5\}$ for adaptation. Notably, due to the triplet loss [29] employed in the base model, which optimizes the distances between samples of the same category, the input must include at least two shots per category. Therefore, for the 1-shot adaptation experiment, we use a single layout clip per category and duplicate it to meet this requirement.

3) *Hyperparameters*: In our experiments, we use SoTA CNN architecture for hotspot detection as proposed in [2] as our base model architecture which incorporates attention modules for enhanced detection accuracy. In our experiments, we set the number of meta-initial training steps to 10,000. Within each training step, the inner loop and outer loop are trained for 1 and 10 iterations, respectively, using randomly sampled layouts with batch sizes of 2 and 8.

B. Generalization Evaluation on Unseen New Designs

We show in Table 2 the generalization performance of our proposed training framework for CNN-based hotspot detection compared against four baselines outlined in Section IV-A2. All experimental results are averaged over 5 runs due to sampling randomness for few-shot adaptation. Our framework demonstrates superior performance across most benchmarks, outperforming baselines in terms of Acc, FAR, and F1 scores. Specifically, we achieve top-1 Acc results in 7 out of the 16 experimental groups and top-2 performance in 15 out of the 16. We note that for those methods surpassing our framework in Acc, they mostly exhibit excessively high FAR ($\geq 27\%$),

compromising their practical applicability. In contrast, our framework maintains a significantly lower FAR ($\leq 6.6\%$) while achieving comparable Acc ($\geq 95.5\%$) in these groups. F1 scores provide comprehensive evaluation of CNN’s detection performance on both hotspots and non-hotspots. Our framework achieves top-1 F1 scores in 14 out of the 16 experimental groups and second place in the remaining two groups. This consistency underscores our framework’s superior generalizability, demonstrating its effectiveness in adapting to previously unseen designs while maintaining exceptional Acc and FAR.

We additionally evaluate the generalization performance of our framework using asynchronous or synchronous meta-learning (denoted as “Ours-”) for meta-initial training. As shown in Table 2, our generalization framework using synchronous meta-learning exhibits either significantly high FAR ($\geq 20.5\%$ in 15 groups) or less accurate Acc (65.3% in the remaining group), indicating its limited capability to generalize to new designs. In contrast, our asynchronous meta-learning method achieves an Acc of at least 95% and a FAR of $\leq 7.5\%$ in 15 out of 16 cases, demonstrating a substantial improvement in generalizability.

C. Generalization Stability Using Different Sampling Methods

As shown by our experimental results in Table 3, the selection of layout clips for few-shot adaptation plays a pivotal role in CNN’s generalization to unseen new designs. We compare the generalization performance of our asynchronous meta-

TABLE 3: Accuracy of various data sampling methods for one-pass few-shot adaptation in generalizing CNN-based lithographic hotspot detection. Acc, FAR, and F1 are averaged over 5 runs for random-based, KM-L, and our sampling methods due to sampling randomness. Red and blue indicate the best-performing means and variances, respectively. Rand: Randomness, Uncert: Probability uncertainty of the meta-initial model, KM-L: layout image-based K -means clustering.

Old Design	New Design	Method	ICCAD 2012						ICCAD 2019					
			1-shot			5-shot			1-shot			5-shot		
			Acc	FAR	F1	Acc	FAR	F1	Acc	FAR	F1	Acc	FAR	F1
B2 & 4	B3	Rand	92.1 ± 1.29	2.1 ± 0.99	93.8 ± 0.84	95.6 ± 3.09	3.4 ± 2.69	94.5 ± 1.31	92.1 ± 4.53	29.8 ± 1.19	73.0 ± 3.00	69.5 ± 31.78	21.7 ± 16.12	41.6 ± 6.37
		Uncert	92.1 ± /	1.2 ± /	94.8 ± /	94.9 ± /	2.1 ± /	95.4 ± /	94.6 ± /	27.5 ± /	69.0 ± /	97.1 ± /	26.2 ± /	49.4 ± /
		KM-L	94.7 ± 0.03	2.0 ± 0.01	95.4 ± 0.02	96.5 ± 0.13	2.6 ± 0.03	95.7 ± 0.08	99.3 ± 0.08	29.9 ± 0.05	76.2 ± 0.14	99.6 ± 0.00	0.4 ± 0.00	98.5 ± 0.00
		Ours	95.5 ± 0.15	2.1 ± 0.04	95.7 ± 0.08	96.5 ± 0.10	2.8 ± 0.05	95.6 ± 0.08	99.6 ± 0.00	31.2 ± 0.06	76.6 ± 0.14	99.7 ± 0.08	0.3 ± 0.08	98.7 ± 0.33
	B5	Rand	96.0 ± 0.55	1.5 ± 0.47	70.3 ± 6.24	96.6 ± 2.27	2.7 ± 2.72	64.4 ± 21.04	98.0 ± 2.74	7.7 ± 2.92	20.8 ± 6.50	91.9 ± 7.60	4.5 ± 3.23	33.2 ± 14.28
		Uncert	95.7 ± /	0.5 ± /	86.0 ± /	96.7 ± /	0.3 ± /	91.5 ± /	100.0 ± /	8.2 ± /	18.6 ± /	94.9 ± /	12.9 ± /	12.7 ± /
		KM-L	97.5 ± 0.16	1.5 ± 0.03	70.9 ± 0.34	97.5 ± 0.16	3.6 ± 0.03	51.0 ± 0.32	100.0 ± 0.00	8.2 ± 0.07	18.5 ± 0.13	100.0 ± 0.00	12.9 ± 0.12	12.7 ± 0.10
		Ours	97.5 ± 0.13	1.5 ± 0.03	70.9 ± 0.33	97.5 ± 0.16	1.6 ± 0.02	69.6 ± 0.20	100.0 ± 0.00	7.0 ± 0.00	21.3 ± 0.00	100.0 ± 0.00	3.3 ± 0.00	35.7 ± 0.00
B3 & 5	B2	Rand	99.4 ± 0.00	4.8 ± 1.04	71.5 ± 7.54	99.2 ± 0.22	2.5 ± 0.75	90.1 ± 2.64	83.1 ± 19.66	5.6 ± 2.46	60.9 ± 5.65	98.4 ± 3.42	15.2 ± 10.48	50.0 ± 16.40
		Uncert	99.4 ± /	10.7 ± /	65.4 ± /	99.2 ± /	6.4 ± /	78.4 ± /	74.0 ± /	6.3 ± /	54.3 ± /	96.6 ± /	5.0 ± /	70.2 ± /
		KM-L	99.4 ± 0.00	4.5 ± 0.02	74.5 ± 0.21	99.4 ± 0.00	2.5 ± 0.01	90.2 ± 0.05	94.7 ± 0.43	7.1 ± 0.03	61.8 ± 0.19	98.9 ± 0.13	10.2 ± 0.05	55.1 ± 0.11
		Ours	99.4 ± 0.00	4.1 ± 0.01	75.3 ± 0.03	99.4 ± 0.00	2.0 ± 0.00	91.9 ± 0.00	97.5 ± 0.43	7.4 ± 0.03	62.2 ± 0.20	97.4 ± 0.20	6.8 ± 0.01	64.1 ± 0.10
	B4	Rand	96.3 ± 0.37	6.8 ± 0.42	58.2 ± 1.47	95.4 ± 0.21	5.3 ± 0.54	64.5 ± 2.32	78.1 ± 15.87	1.9 ± 0.78	60.2 ± 3.69	97.3 ± 5.24	7.4 ± 4.30	42.3 ± 14.96
		Uncert	95.9 ± /	25.4 ± /	28.1 ± /	95.4 ± /	27.1 ± /	26.7 ± /	82.3 ± /	1.7 ± /	64.3 ± /	67.7 ± /	1.1 ± /	63.0 ± /
		KM-L	96.6 ± 0.00	7.8 ± 0.04	55.9 ± 0.12	95.4 ± 0.13	4.8 ± 0.03	66.2 ± 0.18	93.7 ± 0.40	4.3 ± 0.04	49.7 ± 0.14	96.2 ± 0.26	3.7 ± 0.07	54.0 ± 0.49
		Ours	96.6 ± 0.00	6.6 ± 0.04	59.7 ± 0.15	95.4 ± 0.00	4.7 ± 0.03	66.7 ± 0.12	95.0 ± 0.16	4.0 ± 0.03	51.7 ± 0.26	97.6 ± 0.16	4.9 ± 0.09	48.0 ± 0.43

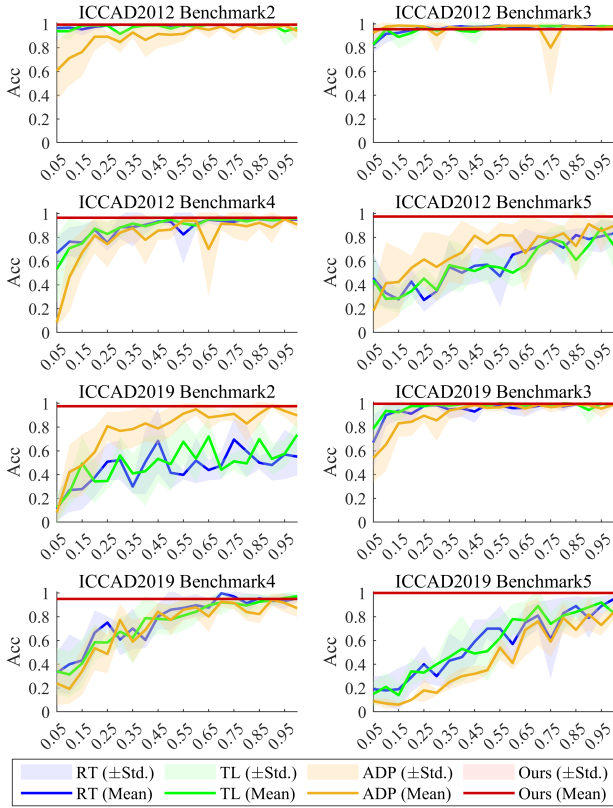


Fig. 4 Comparison of our generalizable framework using one-shot learning with baseline methods using varying data proportions of the new design (x-axis) for one-pass adaptation. The Std. of our framework is ≤ 0.0043 .

learning-based framework using different sampling strategies of new clips, including random sampling (Rand), probability (of the meta-initial model)-based sampling (Uncert), and K -means clustering-based sampling based on either the original layout images (KM-L) or our proposed layout topology-based feature encoding (Ours). All experimental results are averaged over 5 runs for few-shot adaptation and evaluate both the accuracy and stability of generalization on new designs.

Our experimental results reveal that the two K -means-based sampling methods consistently outperform others in Acc and demonstrate superior performance in most evaluations of FAR and F1 scores. Compared to layout image-based sampling, our proposed layout topology-based encoding achieves consistently higher Acc and, in most cases, maintains a competitively extreme-low variance (≤ 0.0043). These findings highlight that our sampling method not only achieves superior generalization performance but also maintains stable results in few-shot sampling and adaptation.

D. Generalization vs. Data Efficiency for Adaptation

We evaluate the data efficiency of our training framework for generalizable CNN-based hotspot detection compared with other baseline solutions using varying numbers of layout clips for adaptation, including retraining (RT), transfer learning (TL), and ADAPT (ADP). These results reflect the amount of new layout clips necessary for the baseline models to achieve comparable generalization accuracy to our framework which uses only one-shot adaptation. As presented in Fig. 4, we report the mean (Mean) and standard deviation (Std.) of the generalization performance of 5 runs using randomly sampled data for adaptation. Our results demonstrate that the baseline models require substantially more layout clips and potentially additional training epochs to reach comparable generalization accuracy of ours. In contrast, our framework achieves the best generalization accuracy and stability using only one-shot adaptation, highlighting its superior generalizability in adapting to new designs.

V. CONCLUSION

In this work, we propose an asynchronous meta-learning-based framework for generalizable DL-based lithographic hotspot detection on unseen new designs using few-shot adaptation. Our proposed layout topology-based sampling strategy for few-shot adaptation enhances training stability for generalization. Our framework enables superior generalization capabilities than prior solutions on unseen new layout designs within the same technology node.

REFERENCES

- [1] Haoyu Yang, Jing Su, Yi Zou, Yuzhe Ma, Bei Yu, and Evangeline FY Young. Layout hotspot detection with feature tensor generation and deep biased learning. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 38(6):1175–1187, 2019.
- [2] Hao Geng, Haoyu Yang, Lu Zhang, Fan Yang, Xuan Zeng, and Bei Yu. Hotspot detection via attention-based deep layout metric learning. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 41(8):2685–2698, 2022.
- [3] Yiyang Jiang, Fan Yang, Bei Yu, Dian Zhou, and Xuan Zeng. Efficient layout hotspot detection via neural architecture search. *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, 27(6):1–16, 2022.
- [4] Hao-Chiang Shao, Guan-Yu Chen, Yu-Hsien Lin, Chia-Wen Lin, Shao-Yun Fang, Pin-Yan Tsai, and Yan-Hsiu Liu. Lithohod: A litho simulator-powered framework for ic layout hotspot detection. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2024.
- [5] FanYang YiyangJiang, Bei Yu, Dian Zhou, and Xuan Zeng. Efficient layout hotspot detection via binarized residual neural network ensemble. *IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS*, 40(7), 2021.
- [6] Shuyuan Sun, Yiyang Jiang, Fan Yang, Bei Yu, and Xuan Zeng. Efficient hotspot detection via graph neural network. In *2022 Design, Automation & Test in Europe Conference & Exhibition (DATE)*, pages 1233–1238. IEEE, 2022.
- [7] Gaurav Rajavendra Reddy, Kareem Madkour, and Yiorgos Makris. Machine learning-based hotspot detection: Fallacies, pitfalls and marching orders. In *2019 IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, pages 1–8. IEEE, 2019.
- [8] Liangjian Wen, Yi Zhu, Lei Ye, Guojin Chen, Bei Yu, Jianzhuang Liu, and Chunjing Xu. Layouttransformer: Generating layout patterns with transformer via sequential pattern modeling. In *Proceedings of the 41st IEEE/ACM international conference on computer-aided design*, pages 1–9. IEEE, 2022.
- [9] Xu He, Yu Deng, Shizhe Zhou, Rui Li, Yao Wang, and Yang Guo. Lithography hotspot detection with fft-based feature extraction and imbalanced learning rate. *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, 25(2):1–21, 2019.
- [10] Wei Ye, Yibo Lin, Meng Li, Qiang Liu, and David Z Pan. Lithoroc: Lithography hotspot detection with explicit roc optimization. In *2019 24th Asia and South Pacific Design Automation Conference (ASP-DAC)*, pages 1–7. IEEE, 2019.
- [11] Haoyu Yang, Luyang Luo, Jing Su, Chenxi Lin, and Bei Yu. Imbalance aware lithography hotspot detection: a deep learning approach. *Journal of Micro/Nanolithography, MEMS, and MOEMS*, 16(3):033504–033504, 2017.
- [12] Jingyu Pan, Chen-Chia Chang, Zhiyao Xie, Jiang Hu, and Yiran Chen. Robustify ml-based lithography hotspot detectors. In *Proceedings of the 41st IEEE/ACM International Conference on Computer-Aided Design*, pages 1–7. IEEE, 2022.
- [13] Kang Liu, Benjamin Tan, Gaurav Rajavendra Reddy, Siddharth Garg, Yiorgos Makris, and Ramesh Karri. Bias busters: Robustifying dl-based lithographic hotspot detectors against backdooring attacks. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 40(10):2077–2089, 2021.
- [14] Kang Liu, Benjamin Tan, Ramesh Karri, and Siddharth Garg. Training data poisoning in ml-cad: Backdooring dl-based lithographic hotspot detectors. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 40(6):1244–1257, 2021.
- [15] Haoyang Sun, Cong Jiang, Xun Ye, Dan Feng, Benjamin Tan, Yuzhe Ma, and Kang Liu. Interpretable cnn-based lithographic hotspot detection through error marker learning. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2024.
- [16] Tao Zhang, Haoyu Yang, Kang Liu, and Zhiyao Xie. Apple: An explainer of ml predictions on circuit layout at the circuit-element level. In *2024 29th Asia and South Pacific Design Automation Conference (ASP-DAC)*, pages 374–379. IEEE, 2024.
- [17] Ran Chen, Wei Zhong, Haoyu Yang, Hao Geng, Fan Yang, Xuan Zeng, and Bei Yu. Faster region-based hotspot detection. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 41(3):669–680, 2020.
- [18] Tianyang Gai, Tong Qu, Shuhan Wang, Xiaojing Su, Renren Xu, Yun Wang, Jing Xue, Yajuan Su, Yayi Wei, and Tianchun Ye. Flexible hotspot detection based on fully convolutional network with transfer learning. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 41(11):4626–4638, 2021.
- [19] Quanshi Zhang, Wenguan Wang, and Song-Chun Zhu. Examining cnn representations with respect to dataset bias. In *Proceedings of the AAAI conference on artificial intelligence*, volume 32, 2018.
- [20] Wei Ye, Mohamed Baker Alawieh, Meng Li, Yibo Lin, and David Z Pan. Litho-gpa: Gaussian process assurance for lithography hotspot detection. In *2019 Design, Automation & Test in Europe Conference & Exhibition (DATE)*, pages 54–59. IEEE, 2019.
- [21] Yifeng Xiao, Miaodi Su, Haoyu Yang, Jianli Chen, Jun Yu, and Bei Yu. Low-cost lithography hotspot detection with active entropy sampling and model calibration. In *2021 58th ACM/IEEE Design Automation Conference (DAC)*, pages 907–912. IEEE, 2021.
- [22] Haoyu Yang, Shuhe Li, Cyrus Tabery, Bingqing Lin, and Bei Yu. Bridging the gap between layout pattern sampling and hotspot detection via batch active learning. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 40(7):1464–1475, 2020.
- [23] Mohamed Baker Alawieh and David Z Pan. Adapt: An adaptive machine learning framework with application to lithography hotspot detection. In *2021 ACM/IEEE 3rd Workshop on Machine Learning for CAD (MLCAD)*, pages 1–6. IEEE, 2021.
- [24] Hassan Gharoun, Fereshteh Momenifar, Fang Chen, and Amir Gandomi. Meta-learning approaches for few-shot learning: A survey of recent advances. *ACM Computing Surveys*, 56(12):1–41, 2024.
- [25] Chelsea Finn, Pieter Abbeel, and Sergey Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In *International conference on machine learning*, pages 1126–1135. PMLR, 2017.
- [26] Allan Gu and Avideh Zakhor. Optical proximity correction with linear regression. *IEEE Transactions on Semiconductor Manufacturing*, 21(2):263–271, 2008.
- [27] Tetsuaki Matsunawa, Jih-Rong Gao, Bei Yu, and David Z Pan. A new lithography hotspot detection framework based on adaboost classifier and simplified feature extraction. In *Design-Process-Technology Co-optimization for Manufacturability IX*, volume 9427, pages 201–211. SPIE, 2015.
- [28] J Andres Torres. Iccad-2012 cad contest in fuzzy pattern matching for physical verification and benchmark suite. In *Proceedings of the International Conference on Computer-Aided Design*, pages 349–350, 2012.
- [29] De Cheng, Yihong Gong, Sanping Zhou, Jinjun Wang, and Nanning Zheng. Person re-identification by multi-channel parts-based cnn with improved triplet loss function. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1335–1344, 2016.